

The PressMate Coffee Press

Final Report

Team Bob the Builders: Will Castro, Jos Moresi, Li Arditi, and Sam Eppley



A compact, steel, single-serve coffee-brewing device

Motivation - Drawbacks of Competitor's Product

1. Hard
to press

2. Need a
separate
cup

3. \$170?!



Play Video

AeroPress Coffee Maker - Steel

Stainless Steel 3 in 1 Coffee Press

\$169.95 **NEW**

A new standard in coffee flavor. Crafted from double-wall stainless steel with a large capacity, our unique 3-in-1 brew method speeds up extraction for less bitterness and a smoother, more full-bodied cup.

Color: **Silver**



4 interest-free installments, or from **\$15.34/mo** with [shop Pay](#)

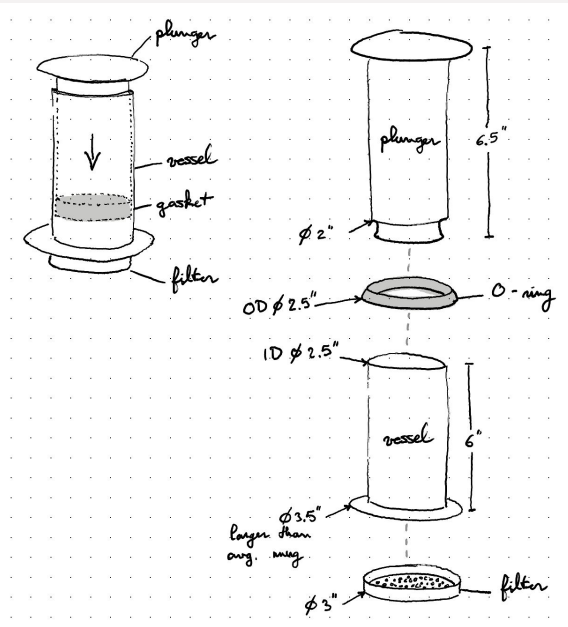
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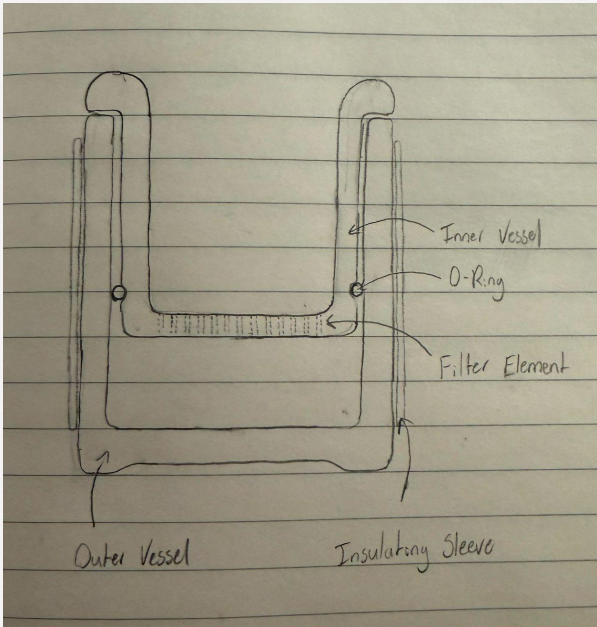
Free shipping over \$55 | Easy Returns

Complete Your Routine

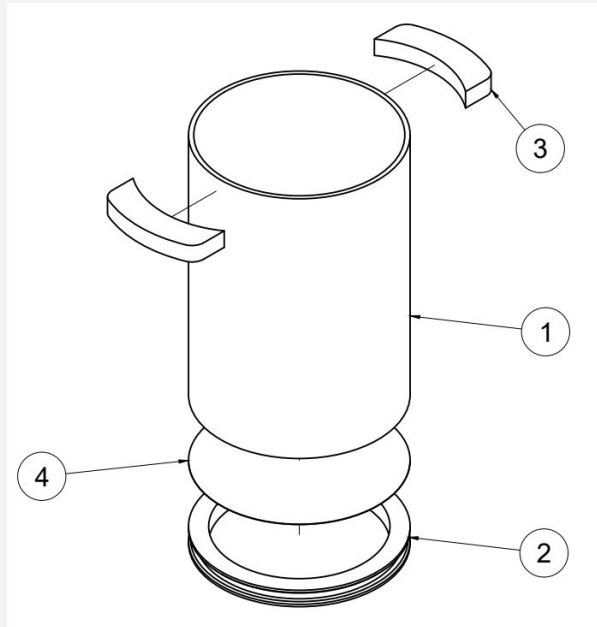
Design Development



Initial Proposal



Revised Proposal



Final Design

Product Concept - Components (PVT version)



Vessel

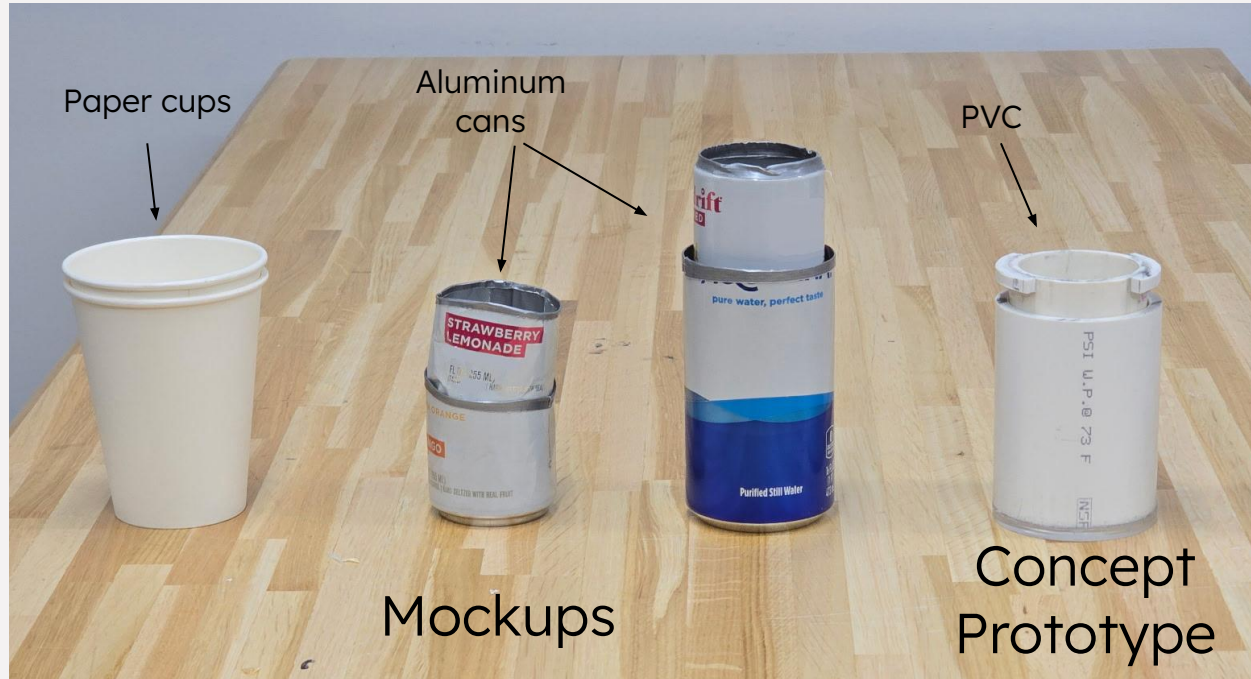


Plunger



Sleeve

Prototype Progression (part 1)



Prototype Progression (part 2)

EVT

Overall product



Plunger and vessel



Components



Sleeve to protect user from hot metal

Lid to protect user from steam

DVT

Plunger and vessel (x3)



Sleeves/coozies (x3)



Components



Seasoned finish for rust resistance

Silicone lip

Lid not needed

PVT

Group picture with all 10 PVT products and tooling



Manufacturing Processes (part 1)

TIG welding vessel and plunger

Jos, our pro welder



Plunger tab welding tool



Sam dremeling down some early learning curve welds



Plunger ring welded to body

Manual and CNC lathe for turning vessel, grooving plunger, and polishing



Manual



CNC

Lathe tooling (inserts to ensure tubes remain round)



Will running the CNC lathe



Li polishing on manual CNC

Manufacturing Processes (part 2)

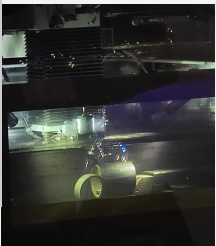
Wire EDM

Wire EDM cutting tube stock to avoid large deformation



Li watching a cut finish

Picture of finished cut taken from outside EPIC windows after hours



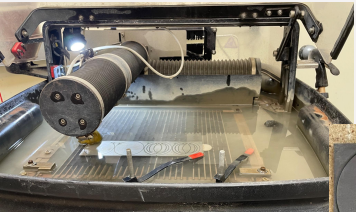
Grapeseed Oil Seasoning

Vessels ready to be baked and plungers in the oven



Waterjet

Cut vessel bottoms, plunger rings, and tabs



Finished cut



Sewing

Inside layer with insulation



Final product



Silicone Dipping

DVT clear lip



PVT black lip curing



A very sophisticated curing rack tool

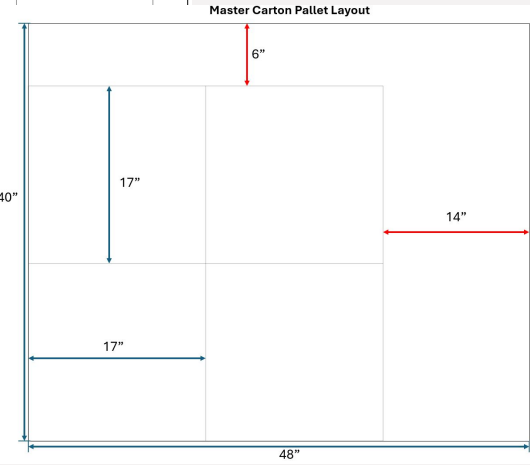
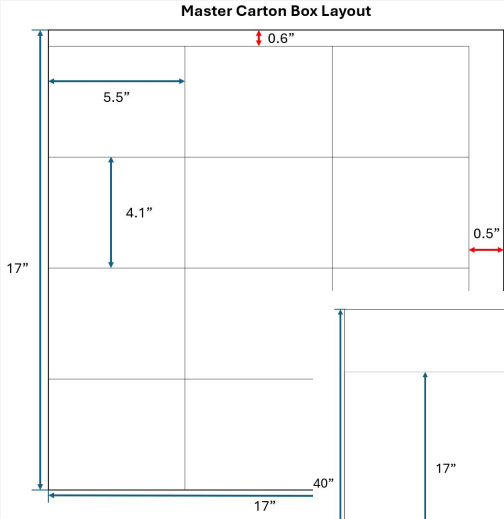
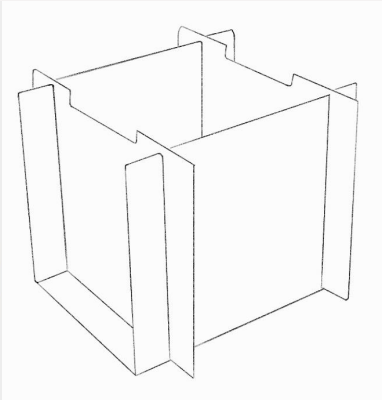
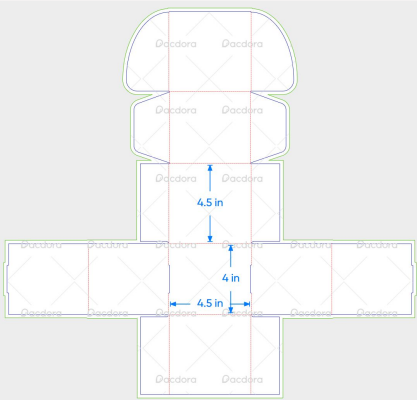


BOM, COGS, Cost

BOM MOCKUP		
	Materials	\$1.00
	Labor	\$39.00
	Total BOM	\$40.00
BOM EVT		
	Materials	\$7.27
	Labor	\$387.27
	Total BOM	\$394.54
BOM DVT		
	Materials	\$22.15
	Labor	\$268.67
	Total BOM	\$290.82
BOM PVT		
	Materials	\$21.52
	Labor	\$277.84
	Total BOM	\$299.36
BOM MP		
	Materials	\$9.55
	Labor	\$1.20
	Total BOM	\$10.75

Packaging

MODEL NO.	INSIDE DIMENSIONS L x W x H	PRICE PER BOX					BUNDLE/ BALE QTY.	LBS./ BN DL.
		25	100	250	500	1,000+		
S-11376	17 x 17 x 10"	\$3.21	\$2.80	\$2.52	\$2.31	\$2.16	25 / 250	42



The box is a Roll End Tuck Front design that is easy to operate, secure, and feels more premium than other box designs.

Left: box rendering, Middle: box template, Right: insert

A 17" x 17" x 10" corrugated box holds 24 PressMates (boxed)

A fully stacked pallet would contain 16 master cartons, for a total 384 PressMates.

Lessons Learned

1. O-Ring dynamics

- O-ring material and surrounding surfaces can have large effect on fit and seal performance

2. Fixturing without deformation

- Keeping tubes round while also being secure on lathes is hard. Any process for the tubes needed to consider possibility of becoming out of round

3. Seasoning

- A little goes a long way. Seasoning technique greatly affects final finish. Very little needs to be used and it must be evenly distributed. Also requires multiple coats, otherwise the finish wears off easily

4. Human Error/Variability

- Even with the same process, same settings, and CNC programs, results vary enough to affect fits between components

5. Have Plan A, Plan B, Plan C, and Plan D. Maybe even Plan E

- As much as you plan, something will happen to throw you off. Make you plans flexible and have backup plans